Women's Specialists of Clear Lake
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Lynch Syndrome Screening

Understanding hereditary cancer risk and the importance of proactive surveillance

WHAT IS LYNCH SYNDROME?

Lynch syndrome (formerly Hereditary Nonpolyposis Colorectal Cancer, or HNPCC) is one of the most common **hereditary cancer syndromes**, affecting approximately **1 in 300 people**. It is caused by a mutation in DNA mismatch repair (MMR) genes.

People with Lynch syndrome have a significantly increased lifetime risk of several cancers — many of which are **highly treatable when caught early** through regular surveillance.

Inheritance Pattern:

- **Autosomal dominant** — one copy of the mutated gene is sufficient
- Each first-degree relative has a 50% chance of inheriting the mutation
- Often runs through multiple generations and both sexes

ASSOCIATED CANCER RISKS

Colorectal Cancer
40–80% lifetime risk
(General pop: 4%)

Endometrial (Uterine)
40–60% lifetime risk
(General pop: 3%)

Ovarian Cancer
10–12% lifetime risk
(General pop: 1.3%)

Gastric Cancer
6–13% lifetime risk
(General pop: <1%)

Urinary Tract
4–12% lifetime risk
Ureter / renal pelvis

Small Bowel
4–7% lifetime risk
Rare in general population

THE 4 MMR GENES

MLH1	MSH2	MSH6	PMS2

Should You Be Tested? Amsterdam Criteria:

- 3 or more relatives with Lynch-associated cancers
- Cancers affecting at least 2 generations
- At least 1 family member diagnosed before age 50
- One case is a first-degree relative of the other two

Universal Tumor Testing:

- All endometrial and colorectal tumors should be tested for MMR status (IHC) — now standard of care
- MSI-High or MMR-deficient result → genetic counseling recommended

SURVEILLANCE RECOMMENDATIONS

If diagnosed with Lynch syndrome:

Colonoscopy	Every 1–2 years starting at age 20–25 (or 2–5 years before earliest family cancer)
Endometrial	Annual endometrial biopsy starting at age 30–35; discuss risk-reducing hysterectomy
Ovarian	Annual pelvic exam + ultrasound + CA-125; discuss BSO after childbearing
Urinary	Annual urinalysis starting at age 30–35 (MSH2/EPCAM carriers)
Risk Reduction	Aspirin use, risk-reducing surgery (hysterectomy + BSO) after childbearing — discuss with Dr. Klein

Refer First-Degree Relatives:

- All first-degree relatives should receive genetic counseling
- Cascade testing can identify affected family members before cancer develops

Knowledge Is Power — Early Detection Saves Lives

Lynch syndrome cancers are among the most treatable when caught early through surveillance. If you have a personal or family history of Lynch-associated cancers, please discuss genetic testing with Dr. Klein. We can refer you to a genetic counselor and create a personalized surveillance plan. Call **(281) 557-0300**
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